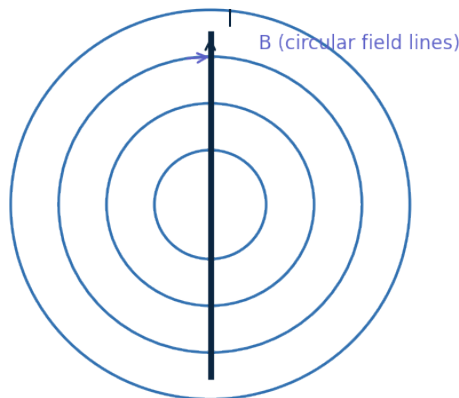


Magnetic Effects of Current

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- A current-carrying conductor produces a magnetic field around it. Direction given by the right-hand thumb rule.
- Biot-Savart Law gives the magnetic field due to a small current element: $dB = (\mu_0/4\pi) I dl \sin(\theta)/r^2$.
- Magnetic field due to a long straight wire at perpendicular distance r : $B = \mu_0 I / (2\pi r)$.
- Magnetic field at the center of a circular current loop of radius R : $B = \mu_0 I / (2R)$.
- Magnetic field inside a long solenoid: $B = \mu_0 n I$, where n is the number of turns per unit length.
- Force on a moving charge in a magnetic field (Lorentz force): $F = qvB \sin(\theta)$, direction given by the right-hand rule ($F = qv \times B$).
- Force on a current-carrying conductor in a magnetic field: $F = BIL \sin(\theta)$.
- Two parallel current-carrying wires attract if currents are in the same direction and repel if in opposite directions.



Circular magnetic field lines around a straight current-carrying conductor.

2. Formula Sheet

Biot-Savart law	$dB = (\mu_0/4\pi) I dl \sin(\theta) / r^2$
Field due to straight wire	$B = \mu_0 I / (2 \pi r)$
Field at center of circular loop	$B = \mu_0 I / (2R)$
Field inside a solenoid	$B = \mu_0 n I$
Lorentz force (moving charge)	$F = qvB \sin(\theta)$
Force on current-carrying wire	$F = BIL \sin(\theta)$
Force between two parallel wires (per unit length)	$F/L = \mu_0 I_1 I_2 / (2 \pi d)$
Radius of circular path of a charged particle in B	$r = mv / (qB)$

3. Practice Problems (22 questions)

- Find the magnetic field at a distance of 0.1 m from a long straight wire carrying a current of 5 A ($\mu_0 = 4 \pi \times 10^{-7} \text{ T m/A}$).
- Find the magnetic field at the center of a circular loop of radius 0.05 m carrying a current of 2 A.
- A solenoid has 1000 turns per meter and carries a current of 3 A. Find the magnetic field inside it.
- A charged particle with charge $2 \times 10^{-6} \text{ C}$ moves at $3 \times 10^5 \text{ m/s}$ perpendicular to a magnetic field of 0.5 T. Find the force on it.
- A straight wire of length 0.5 m carries a current of 4 A in a magnetic field of 0.2 T, perpendicular to the wire. Find the force on the wire.
- State the right-hand thumb rule for finding the direction of the magnetic field due to a current-carrying wire.
- Two parallel wires, 0.1 m apart, carry currents of 5 A and 10 A in the same direction. Find the force per unit length between them.
- A proton (charge $1.6 \times 10^{-19} \text{ C}$, mass $1.67 \times 10^{-27} \text{ kg}$) moves in a circular path of radius 0.2 m in a magnetic field of 0.5 T. Find its speed.
- Explain why a current-carrying loop behaves like a magnetic dipole, and define its magnetic moment.
- State whether two parallel wires carrying current in opposite directions attract or repel each other, and explain why.
- Find the magnetic field 0.02 m from a long straight wire carrying a current of 10 A.
- A cyclotron uses a magnetic field to accelerate charged particles in a circular path. Explain briefly the basic principle behind its working.
- An electron moves parallel to a magnetic field. Find the force experienced by it. Explain your reasoning.
- Find the magnetic moment of a current loop carrying 2 A of current with an area of 0.01 m^2 .

15. A wire carrying current 3 A is placed parallel to a magnetic field of 0.4 T. Find the force per unit length on the wire.
16. State the SI unit of magnetic field and define it in terms of force on a moving charge.
17. Explain how a moving-coil galvanometer uses the magnetic effect of current to detect small currents.
18. A charged particle enters a magnetic field at an angle of 90 degrees to it. Describe the shape of its path and explain why.
19. Find the magnetic field at the center of a circular coil of 50 turns, radius 0.1 m, carrying a current of 1 A.
20. State Ampere's circuital law and give one situation where it simplifies calculating the magnetic field.
21. Two long straight parallel wires carry equal currents of 3 A in opposite directions, separated by 0.05 m. Find the force per unit length between them and state whether it is attractive or repulsive.
22. A charged particle moves in a straight line through a region containing both electric and magnetic fields. What can you conclude about the relationship between the electric and magnetic forces on it?

Answer Key

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|---|---|
| 1. $B = 1.0 \times 10^{-5} \text{ T}$ | 2. $B = 2.51 \times 10^{-5} \text{ T}$ (approx) |
| 3. $B = 3.77 \times 10^{-3} \text{ T}$ (approx) | 4. $F = 0.3 \text{ N}$ |
| 5. $F = 0.4 \text{ N}$ | 6. Conceptual - thumb points along current, curled fingers show field direction |
| 7. $F/L = 1.0 \times 10^{-4} \text{ N/m}$ | 8. $v \approx 2.99 \times 10^7 \text{ m/s}$ |
| 9. Conceptual - loop's field pattern resembles a bar magnet; magnetic moment $m = NIA$ | 10. Conceptual - repel each other (opposite currents produce opposing fields between them) |
| 11. $B = 1.0 \times 10^{-4} \text{ T}$ | 12. Conceptual - magnetic field keeps particle in a circular path while an alternating electric field accelerates it across the gap each half cycle |
| 13. $F = 0$ (force is zero when velocity is parallel to B , since $\sin(0) = 0$) | 14. $m = 0.02 \text{ A m}^2$ |
| 15. $F/L = 1.2 \text{ N/m}$ | 16. Conceptual - tesla (T); 1 T is the field producing 1 N of force on a 1 C charge moving at 1 m/s perpendicular to it |
| 17. Conceptual - current-carrying coil in a magnetic field experiences a torque proportional to current, causing pointer deflection | 18. Conceptual - circular path, since the magnetic force is always perpendicular to velocity, providing centripetal force |
| 19. $B \approx 3.14 \times 10^{-4} \text{ T}$ | 20. Conceptual - useful for finding B due to a long straight wire, solenoid, or toroid using symmetry |
| 21. $F/L = 3.6 \times 10^{-5} \text{ N/m}$, repulsive (opposite currents repel) | 22. Conceptual - electric and magnetic forces on the particle must be equal and opposite (net force zero) for it to continue in a straight line |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.